

ATTACHMENT # 4 -STATEMENT OF WORK

Upgrade Boiler - Add magnetic separator, repair floor trip hazards and replace hot water manifold.

1. INTRODUCTION / SUMMARY OF WORK PROJECT

This Scope of Work (SOW) defines the requirements for:

- **Installation of magnetic separator**
Supply and install a suitable filter in the designated location to improve system performance and ensure compliance with operational standards.
- **Repair of Floor Trip Hazards:**
Identify and repair all floor areas presenting trip hazards. Work includes leveling, resurfacing, or replacing flooring materials as necessary to eliminate safety risks and ensure a smooth, hazard-free surface.
- **Replacement of Hot Water Manifold:**
Remove the existing hot water manifold and supply and install a new manifold. Ensure all connections are secure and the system is fully operational upon completion.

All work will be performed in accordance with applicable safety and quality standards. The area will be left clean and free of debris upon completion.

The project will be located at:

Calle de Serrano, 75, 28006 Madrid, Spain

2. SCOPE OF WORK

2.1 PRIOR TO IMPLEMENTATION

Required site visit and survey. The contractor shall be responsible for researching the existing conditions and matters that affect the cost or performance of the services. Contractor is encouraged to attend a site visit to confirm all measurements and survey existing layout prior to submission of BOM and quotation. Measurements provided are approximate and US Embassy shall not be held responsible for inaccurate dimensions. The contractor shall coordinate and obtain any permit required by local authorities for the execution of the project.

DOCUMENT SUBMITTAL PACKAGE. Submit before awarding process, document submittal package that includes:

- Product Data: Manufacturer's data sheets on each product to be used, including:
- Submit manufacturer's installation instructions, including preparation, storage, and handling requirements.

- Complete finalized Work Plan including Critical Path Method (CPM) schedule.

DOCUMENT SUBMITTAL PACKAGE. Submit to the CO and/or COR within 15 days of Notice to Proceed, document submittal package that includes:

Activity Hazard Analysis (AHA) with detail specifications of the safety plan to be implemented.

2.2 IMPLEMENTATION

AREA ISOLATION AND ACCESS RESTRICTION

The contractor shall provide a plan for the isolation and restriction of access to the area to be intervened.

Works must only be carried out by staff with technical knowledge to do so and always complying with the security rules.

Protection and isolation of the area. Please include water protection like provisional roof in case of rain.

2.2.1 PROJECT DETAILS

2.2.1.1 Supply and Installation of magnetic separator

Spirocross XC100 FM or similar degasser and Magnetic Separator: Supply and installation, per unit, of degassers and magnetic particle separators, manufactured in DN100, installed on the return branches of the primary production groups of the boiler and the rooftop gas boilers (Adisa).

2.2.1.2 Repair of Floor Trip Hazards: Epoxy Resin Flooring (Square meter of epoxy resin floor coating, with anti-slip treatment and finish in RAL color of choice.)

Surface Preparation

Thoroughly clean the substrate to remove dust, grease, oil, and any loose material. Repair cracks, holes, or uneven areas using appropriate fillers or repair mortars. Mechanically prepare the surface (e.g., by grinding or shot blasting) to ensure good adhesion of the resin. Ensure the surface is dry and free from contaminants.

Priming

Apply a suitable primer to the prepared surface using a roller or brush. Allow the primer to cure as per the manufacturer's instructions. This step promotes adhesion and seals the substrate.

Mixing the Resin

Prepare the resin components (base and hardener) according to the manufacturer's instructions. Mix thoroughly using a mechanical mixer to achieve a uniform consistency.

Application of Resin Layer

Pour the mixed resin onto the primed surface. Spread the resin evenly using a notched trowel, roller, or squeegee to the specified thickness. For anti-slip finishes, broadcast aggregate (such as quartz sand) onto the wet resin as required.

Finishing

If an anti-slip or decorative finish is required, apply a topcoat or sealant after the base layer has cured. Remove any excess aggregate before applying the topcoat.

Curing

Allow the resin floor to be cured for the time specified by the manufacturer, ensuring the area is protected from traffic, dust, and moisture during this period.

Inspection and Handover

Inspect the finished floor for uniformity, adhesion, and finish quality. Clean the area and remove any debris or masking materials. Handover the completed resin floor, providing maintenance and care instructions as needed.

2.2.1.3 Replacement of Hot Water Manifold

Dismantling of Existing Pipes and Elements

Controlled dismantling and removal of the existing room, including cutting of pipes and removal of pumps and manifolds, as well as disposal of the removed elements at a landfill.

Temporary Domestic Hot Water (DHW) Production Heat Pump

Rental and installation of a heat pump system for temporary DHW production during the remodeling works in the room. Installed indoors and near the storage tanks, complete with piping, valves, and temporary electrical connection.

Fabrication and Installation of Auxiliary Bypass for DHW and Pool-Cafeteria Circuits

Fabrication and installation of an auxiliary bypass for the DHW and Pool-Cafeteria circuits, installed directly on the primary branch of the boiler in the machine room, complete with tapping points and piping for installation. Operated manually in case of malfunction. Does not pass through the main heating manifold.

Reconnection of DHW System Circuit

Unit for reconnection and installation of new piping and the existing DHW heat exchanger, including shut-off valves, fittings, and insulation. Executed in AISI 316L stainless steel, including bypass between tanks.

DHW Return Pump

Supply and installation of a single water pump for the DHW return circuit, Grundfos

model MAGNA1 25-120N, including fittings, filter, and shut-off valves. Installed and operational. **Two filters are required: One per boiler.**

2.2.2 AFTER IMPLEMENTATION

Detail engineering and "as-built" documents: After final completion of the work, but before final acceptance thereof, the Contractor shall provide: a complete set of "as-built" drawings, based upon the record set of drawings, marked to show the details of construction as actually accomplished.

When all site work is complete, the Contractor's site superintendent shall review all activities and walk the project area with the COR to ensure that the embassy staff is satisfied, and all work items are acceptable. If any items are not completed, they shall be noted and provided to the Contractor to be addressed and completed immediately.

The Contractor shall ensure that, upon demobilization, the site is left in a clean and orderly appearance, equal to or better than prior to start of work.

3. CODES AND STANDARD

Any design specifications shall conform to the more stringent requirements of the following attached codes and standards.

Occupational Safety & Health Administration Codes (OSHA)
ACE EM 385
International Building Code (ICBO)
International Mechanical Code (ICBO).
Mechanical Systems - ASHRAE and SMACNA Standards.
National Fire Protection Association (NFPA 101) Life Safety Code.

4. PERIOD OF PERFORMANCE

The Contractor shall be required to:

- (a) commence work under this contract within ten (10) calendar days after the date the Contractor receives the notice to proceed,
- (b) prosecute the work diligently, and,
- (c) complete the entire work ready for use not later than one hundred sixty days (60) calendar days** after Notice to Proceed.

** Contractor might be required to accomplish work in two phases based on CLIN no. 1, 2 and 3 above. If that is the case, the Government will issue two different NTP and contractor will be required to complete works no later than 60 days after Notice to Proceed (35 days of actual work for CLINS 1 and 2 and 35 days for CLIN 3)

The time stated for completion shall include final cleanup of the premises.